
Sync2Act

Mixed-Modality Data Corruption Study

Full datasets, six ablations, three random seeds

Experiment size	3 datasets $\times$ 6 ablations $\times$ 5 conditions $\times$ 3 seeds
Completion	270 / 270 runs completed and passed integrity audits
Training scope	Full episodes, all frames and cameras; training-only corruption
Main metrics	Clean-test action MSE and ratios to the same model/seed's clean-training baseline
Evaluation	Paired first-step and causal-ensemble evaluation on identical models and test data
Report date	2026-10-02

Findings at a glance

Under first-step evaluation, for shifted or missing action labels, correctly aligned Oracle label quality nearly removes average degradation: Quality-Full has relative MSE $0.977\times$, versus ACT-Lite's $2.432\times$. Under mixed damage, the respective ratios are $1.144\times$ and $1.535\times$. State-quality input also helps, while image quality has no clear benefit under the current offline metric.

Temporal-ensemble evaluation: Both action-selection methods use the same models and test data. The report distinguishes ratios to clean training from actual MSE changes between methods.

1 Research question and evidence boundaries

This study examines four mixed-quality training conditions: image damage, state damage, action-label damage, and a mixture across modalities. When clean and damaged segments coexist, should modality-specific quality enter the model input, weight the loss, or do both?

Quality is **Oracle quality** supplied by the known synthetic corruption process. It encodes known fault provenance using heuristic scores, not calibrated correctness probabilities, a proven performance upper bound, or automatically available deployment signals. All results measure **offline action prediction** on clean test data. MSE and MAE do not substitute for closed-loop success, return, or safety-failure rates in simulation or hardware.

Two offline evaluation methods are compared: first-step prediction selects each chunk’s first action; causal temporal ensembling fuses overlapping predictions for the same target time. Both use identical checkpoints, test episodes, and normalization. Section 6 presents overall curves and paired actual MSE changes. Each degradation curve uses the clean-training baseline evaluated by the same method; before/after changes use first-step MSE as their reference.

2 Datasets and training protocol

The three robot datasets passed the single-task check. Training uses all recorded camera views in each dataset. A fixed episode-level split precedes training-only corruption, avoiding adjacent-frame leakage between training and testing.

Dataset	Episode	Frame	Cameras	State / Action	Train / Val / Test
XArm Lift Medium	800	20,000	1	4 / 4	640 / 80 / 80
Koch Pick Place 1 Lego	100	32,951	2	6 / 6	80 / 10 / 10
ALOHA Static Battery	49	29,400	4	14 / 14	41 / 4 / 4

Table 1: Full-dataset statistics. Training uses one, two, and four recorded camera views, respectively.

Shared settings: 10 epochs, CUDA, batch size 256, segment length 16, training seeds 7/17/27, and split seed 7. The image longest edge is resized to 64 pixels. Normalization statistics are computed only from clean training episodes and frozen across all 90 combinations within a dataset. Validation and test remain clean.

Data separation and audits

All 270 runs retain individual checkpoints, prediction CSVs, per-episode metrics, and manifests. Audits confirm matching checkpoint/quality schemas, experiment signatures, and run metadata; one normalization definition per dataset; and no missing or duplicate combinations.

3 Mixed-corruption conditions

Each episode is divided into contiguous segments, each assigned exclusively to one component according to mixture weights. A segment is not counted in multiple percentage buckets. Shifts use two frames with invalid boundaries marked missing. Modality missingness sets the corresponding input/label and quality to zero, and records missing masks and provenance.

Training condition	Segment composition
Clean	100% clean
Mixed image damaged	50% clean + 40% image shift 2 + 10% image missing
Mixed state damaged	50% clean + 40% state shift 2 + 10% state missing
Mixed action damaged	50% clean + 40% action shift 2 + 10% action-label missing
Mixed three corruptions	20% clean + 20% image shift 2 + 20% state shift 2 + 25% action shift 2 + image/state/action missing, each 5%

Table 2: The five formal training conditions.

Across datasets and seeds, segment proportions differ from targets by at most approximately 0.03 percentage points. Final episode segments can have fewer than 16 frames, so frame proportions differ slightly: by at most about 0.35 percentage points for single-modality conditions and 0.83 for the mixed condition.

4 Six ACT / quality ablations

Model	Quality input	Label-weighted loss	Purpose
ACT-Lite	No	No	Baseline without quality
Quality-Input	Yes	No	Input conditioning only
Quality-Weighted-Loss	No	Yes	Label downweighting only
Quality-Full	Yes	Yes	Full method
Quality-Shuffled	Shuffled	Yes	Break sample correspondence
Quality-Constant	Constant	Yes	Mark every sample as clean

Image/state missingness does not remove whole samples. Samples remain in training, and observation quality/missing masks describe unreliable inputs. Missing actions concern label reliability: only label-quality-weighted loss reduces or removes their supervision contribution.

5 Metrics and aggregation

The primary metric is action mean squared error on the clean test set:

$$\text{MSE} = \frac{1}{Nd} \sum_{i=1}^N \|\hat{a}_i - a_i\|_2^2.$$

Action units and scales differ across datasets. Cross-dataset aggregation therefore uses ratios within the same model and seed:

$$R = \frac{\text{MSE}_{\text{damaged train} \rightarrow \text{clean test}}}{\text{MSE}_{\text{clean train} \rightarrow \text{clean test}}}.$$

$R = 1$ matches clean training; $R > 1$ indicates degradation. Overall results first compute R within each dataset/model/seed, then average nine ratios across three datasets and three seeds.

6 Overall results

6.1 Overall results with first-step prediction

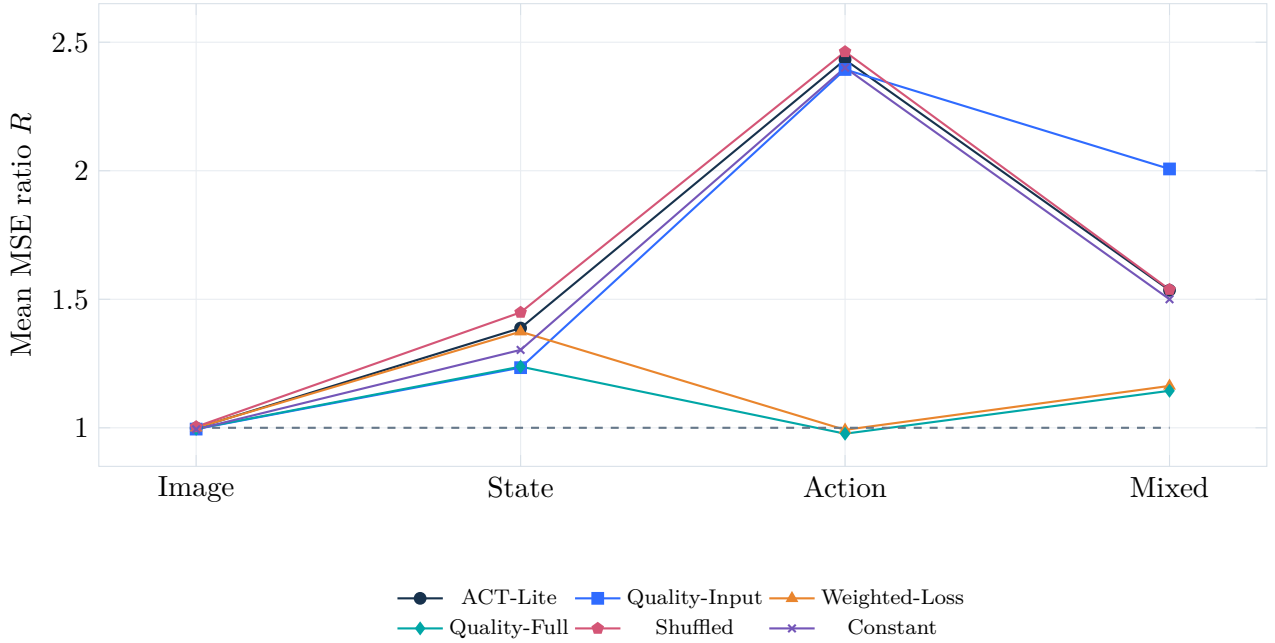


Figure 1: Mean relative MSE across three datasets and three seeds. Lower is better; the dashed line denotes clean training.

Model	Image damaged	State damaged	Action damaged	Mixed
ACT-Lite	0.999	1.388	2.432	1.535
Quality-Input	0.995	1.234	2.394	2.007
Weighted-Loss	1.000	1.374	0.992	1.163
Quality-Full	0.996	1.238	0.977	1.144
Shuffled	1.004	1.449	2.463	1.538
Constant	0.994	1.303	2.401	1.500

Table 3: Mean MSE ratios. Green marks column minima, not statistical significance.

6.2 Overall results with temporal ensembling

Both evaluation methods use the same 270 checkpoints, test episodes, and normalization statistics; decay=0.7 was fixed before evaluation. Only predictions from chunks starting at or before the target time are fused; history resets at episode boundaries. First-step prediction consistency audits: 270/270 passed. Both evaluation methods use the same normalization rule: within each dataset/model/seed, divide the condition’s test MSE by that model and seed’s clean-training test MSE, then average nine ratios. The ensemble curve uses ensembled MSE in both numerator and denominator. Categories, six-model colors, and vertical ranges match across methods.

$$R_{\text{ens}} = \frac{\text{MSE}_{\text{damaged train, ens}}}{\text{MSE}_{\text{clean train, ens}}}.$$

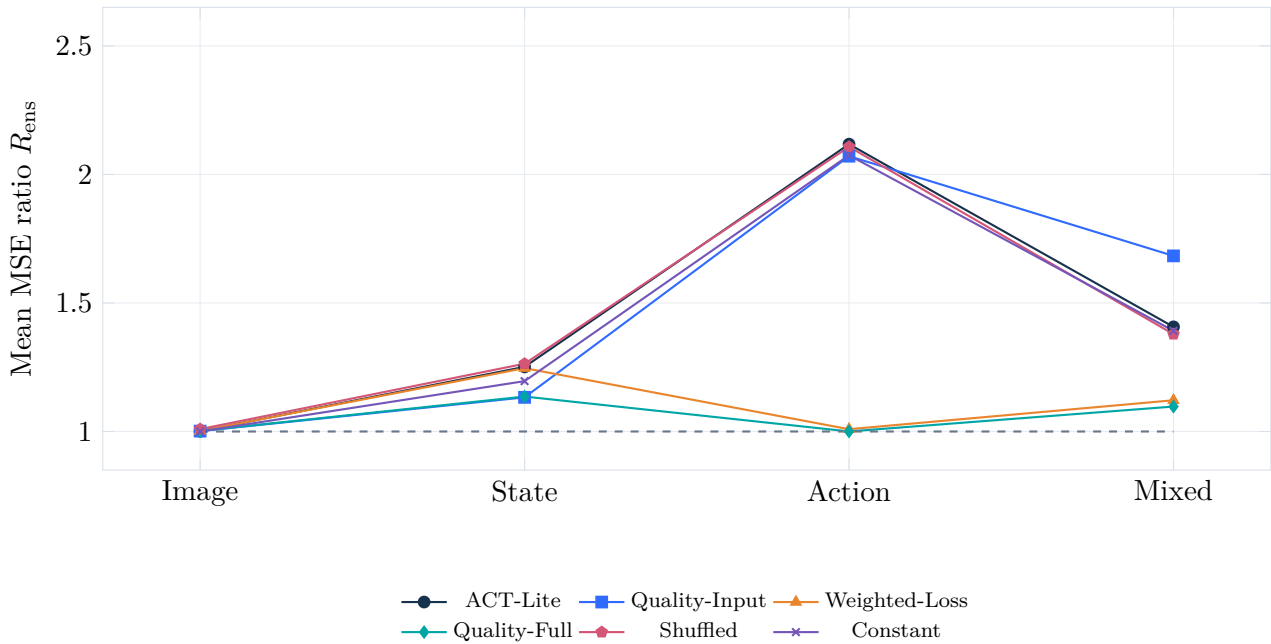


Figure 2: Mean relative MSE with temporal ensembling. Each point averages nine ratios (three datasets, three seeds); the dashed line is the ensembled clean-training baseline.

Model	Image	State	Action	Mixed
ACT-Lite	1.0012	1.2514	2.1177	1.4070
Quality-Input	1.0015	1.1328	2.0711	1.6831
Weighted-Loss	1.0018	1.2465	1.0089	1.1215
Quality-Full	1.0022	1.1360	1.0002	1.0970
Shuffled	1.0091	1.2636	2.1074	1.3787
Constant	0.9999	1.1959	2.0747	1.3916

Table 4: Mean ensemble MSE ratios measure degradation relative to clean training. Paired comparisons report actual error changes between methods.

6.3 MSE before and after ensembling

Compute 100 times (ensemble MSE / first-step MSE - 1) per run, then average over three datasets and three seeds. Negative values mean lower MSE; positive values mean higher MSE. Raw MSE is not averaged across robots.

Model	Clean	Image	State	Action	Mixed
ACT-Lite	+27.37%	+27.72%	+15.23%	+11.47%	+16.74%
Quality-Input	+27.65%	+28.58%	+17.27%	+11.29%	+8.59%
Weighted-Loss	+27.32%	+27.68%	+15.87%	+29.56%	+22.52%
Quality-Full	+27.67%	+28.46%	+17.22%	+31.16%	+22.64%
Shuffled	+27.51%	+28.10%	+11.55%	+10.29%	+14.60%
Constant	+27.58%	+28.29%	+17.16%	+11.10%	+17.89%

Table 5: Mean paired MSE changes. Each cell averages nine runs; positive values mean increased error.

Each row below averages 15 runs within one dataset/model: five training conditions and three seeds. MSE difference = ensemble mean - first-step mean; change (%) = 100 times (ensemble mean / first-step mean - 1). This ratio of means differs from the mean of per-run percentages above.

Dataset	Model	First-step	Ensemble	Difference	Change
XArm	ACT-Lite	0.39085	0.39331	+0.002467	+0.63%
XArm	Quality-Input	0.39089	0.39226	+0.001367	+0.35%
XArm	Weighted-Loss	0.38469	0.38938	+0.004692	+1.22%
XArm	Quality-Full	0.38362	0.38759	+0.003975	+1.04%
XArm	Shuffled	0.39163	0.39363	+0.001998	+0.51%
XArm	Constant	0.39087	0.39338	+0.002507	+0.64%
Koch	ACT-Lite	11.986	16.958	+4.972	+41.48%
Koch	Quality-Input	13.35	18.178	+4.828	+36.16%
Koch	Weighted-Loss	8.224	12.919	+4.695	+57.08%
Koch	Quality-Full	8.1593	12.941	+4.781	+58.60%
Koch	Shuffled	12.532	17.4	+4.867	+38.84%
Koch	Constant	11.936	16.915	+4.978	+41.71%
ALOHA	ACT-Lite	0.0004896	0.00053513	+4.553e-05	+9.30%
ALOHA	Quality-Input	0.00050224	0.00054808	+4.584e-05	+9.13%
ALOHA	Weighted-Loss	0.00033546	0.00037695	+4.149e-05	+12.37%
ALOHA	Quality-Full	0.00031741	0.00036239	+4.498e-05	+14.17%
ALOHA	Shuffled	0.00050376	0.00054189	+3.813e-05	+7.57%
ALOHA	Constant	0.00048196	0.00053488	+5.292e-05	+10.98%

Table 6: Within-dataset actual MSE comparison. Each row averages five conditions and three seeds (15 runs).

270 runs: mean per-run relative MSE change = +21.20%. A lower relative-degradation curve does not imply lower actual MSE: ensembling also changes the clean-training denominator. Assess accuracy using actual first-step and ensemble MSE together. These results concern offline action prediction only. Full condition comparisons are in the expandable HTML table and `temporal_ensemble/mse_by_dataset_model_condition.csv`.

Reproduce: `python tools/evaluate_temporal_study.py --device cuda --decay 0.7 --report-only`.

7 Interpretation by fault modality

This section and Section 8 analyze modalities and paired models using first-step metrics. Section 6 reports ensemble results and MSE comparisons between evaluation methods.

7.1 Action-label damage: weighted loss is central

Under action-label damage, ACT-Lite degrades to $2.432\times$ and Quality-Input remains at $2.394\times$. Quality-Input conditions on observation quality but does not receive action-label quality. Bad labels still enter its ordinary loss at full weight.

Quality-Weighted-Loss reaches $0.992\times$ and Quality-Full $0.977\times$, while Shuffled and Constant reach $2.463\times$ and $2.401\times$. These controls support the importance of **correctly aligned per-sample label quality**, rather than attributing gains solely to extra inputs, random downweighting, or capacity.

Model	XArm MSE	Koch MSE	ALOHA MSE
ACT-Lite	0.406790	21.5804	0.0009031
Quality-Input	0.405661	21.4460	0.0009163
Weighted-Loss	0.388007	7.3369	0.0002718
Quality-Full	0.387742	7.3168	0.0002662
Shuffled	0.407930	21.8305	0.0009603
Constant	0.406804	21.5116	0.0009184

Table 7: Three-seed mean absolute MSE under action-label damage. Magnitudes must not be compared across datasets.

7.2 State damage: quality input helps partially

State damage raises ACT-Lite to $1.388\times$. Quality-Input and Full reach $1.234\times$ and $1.238\times$, versus Weighted-Loss at $1.374\times$ without observation quality. Both remain above one: quality metadata identifies unreliable state but does not restore shifted or zeroed information.

7.3 Image damage: limited sensitivity of the offline metric

All models remain near $1.0\times$ under image damage, without consistent separation between correct, shuffled, and constant quality. At the current 64-pixel setting, tasks, and offline MSE, 40% two-frame image shifts plus 10% all-camera missingness do not produce clear average degradation. This does not establish that closed-loop control is insensitive to visual faults.

7.4 Mixed damage: gains mainly from handling bad labels

Under mixed damage, ACT-Lite reaches $1.535\times$, Input $2.007\times$, Weighted-Loss $1.163\times$, and Full $1.144\times$. Full is only slightly better than Weighted-Loss; the main benefit is label weighting, with smaller, seed-dependent additional contributions from observation quality.

8 Dataset differences and paired comparisons

Dataset	Model	Image	State	Action	Mixed
XArm Lift	ACT-Lite	1.000	1.024	1.073	1.059
	Quality-Full	1.000	1.013	1.024	1.029
Koch Pick Place	ACT-Lite	1.000	1.467	2.958	1.774
	Quality-Full	1.017	1.270	0.964	1.123
ALOHA Battery	ACT-Lite	0.997	1.674	3.265	1.773
	Quality-Full	0.973	1.430	0.942	1.279

Table 8: Three-seed mean MSE ratios by dataset.

XArm is less sensitive overall; Koch and ALOHA show larger state/action effects, motivating multi-dataset evaluation. Full has lower absolute MSE than ACT-Lite in 31/36 damaged pairs and lower own-baseline-normalized ratios in 31/36. Absolute MSE is lower in 32/36 comparisons with Shuffled and 29/36 with Constant.

Comparison	Full: lower MSE	Full: lower ratio	Pairs
Quality-Full vs ACT-Lite	31	31	36
Quality-Full vs Shuffled	32	31	36
Quality-Full vs Constant	29	29	36
Quality-Full vs Quality-Input	29	28	36
Quality-Full vs Weighted-Loss	23	24	36

9 Conclusions and next steps

Main findings

1. **Oracle label quality is effective.** It substantially mitigates action shift/missingness on Koch and ALOHA; shuffled/constant controls support the importance of alignment.
2. **State-quality input provides moderate gains.** Average state-damage error falls, but missing information is not recovered and results remain worse than clean.
3. **Image quality has no clear benefit here.** Current severity, resolution, and offline MSE may not reveal closed-loop visual risk.
4. **Full has the lowest aggregate mixed-damage ratio.** Its margin over Weighted-Loss is small; label weighting remains the primary contribution.
5. **Ensembling at fixed decay=0.7 does not reduce overall actual MSE.** The mean per-run MSE percentage change across 270 runs is +21.20%. Lower degradation ratios do not establish lower actual error or closed-loop benefits.

Recommended next-stage sequence:

1. Train automatic image/state/action-label quality estimators from known corruption provenance and report calibration error;
2. Replace Oracle quality with predictions and quantify sensitivity to estimation errors;
3. Increase image-shift/missingness severity and duration, retain higher-resolution features, and test whether current near-neutral results reflect limited sensitivity;
4. Run closed-loop simulation or robot rollouts, reporting success, safety-failure types, and recovery;
5. If closed-loop trends persist, expand seeds, tasks, and severities, using paired confidence intervals or significance tests.

10 Reproducibility and artifact index

Recorded training-function wall time on the GPU path totals 5.40 hours, excluding external download, four-camera video decoding, final evaluation, and reporting; in-loop validation/checkpoint overhead remains included. Integrity checks passed for 270/270 runs, with checkpoint metadata matching manifests. Old experiment directories were not used for resume or aggregation.

Content	Path
Study definition	runs/mixed_modality_damage_v1_3_0/study.json
Structured summary	runs/mixed_modality_damage_v1_3_0/results.csv
Full JSON evidence	runs/mixed_modality_damage_v1_3_0/results.json
Interactive HTML report	runs/mixed_modality_damage_v1_3_0/report_en.html
Per-run records	runs/mixed_modality_damage_v1_3_0/<dataset>/<model>/<condition>/seed-*/

Generated from measured outputs using experiment schema v3 and quality schema v2.

Training version: Sync2Act v1.2.0; Git commit a5b4fbbe1e7cb8c9704c3dbfc19a1ba563662068.

Temporal-evaluation source fingerprints, environment, and protocol: temporal_ensemble/summary.json.